

# Notes: Lagerborg, Pappa, and Ravn (RES, 2023)

## Sentimental Business Cycles

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# 1 Big picture

- **Question.** Do autonomous shifts in consumer sentiment have causal macroeconomic effects, or are survey-based confidence measures merely noisy reflections of fundamentals?
- **Central empirical problem.** Consumer confidence comoves strongly with real activity, but this does not by itself identify sentiment shocks. Confidence may fall because households observe weak fundamentals, or fundamentals may weaken because sentiment falls.
- **Identification strategy.** The paper identifies autonomous sentiment shocks using fatalities in mass shootings as an external instrument for innovations in consumer expectations. The identifying idea is that mass shootings are salient bad news that affect psychological well-being and confidence but are plausibly unrelated to macroeconomic fundamentals.
- **Main confidence variable.** The benchmark confidence measure is the University of Michigan Index of Consumer Expectations (ICE), which focuses on households' forward-looking views about personal finances and the aggregate economy.
- **Main sample.** The benchmark monthly sample runs from January 1965 to August 2007. The authors also examine a longer sample through November 2018.
- **Benchmark method.** The paper estimates a proxy SVAR with mass-shooting fatalities as an external instrument for the structural sentiment shock.
- **Main result.** A negative sentiment shock causes a persistent decline in consumer confidence and triggers a recessionary response: industrial production falls, unemployment rises, labour market tightness deteriorates, and private consumption declines.
- **Nominal side.** The shock has much less obvious nominal effects. CPI effects are modest, and the federal funds rate declines with a lag.
- **Stock market and uncertainty.** Real stock prices fall persistently; macro uncertainty rises initially but is not the main driver of the identified shock.
- **Micro validation.** Using individual survey data and county-level variation in mass shootings, the paper shows that respondents living near mass-shooting counties experience significant declines in confidence.
- **Robustness.** The findings survive alternative instrument definitions, longer samples, placebo reshuffling of shooting dates, local projections, and comparison with Cholesky-identified confidence shocks.
- **Variance contribution.** Sentiment shocks explain a non-negligible fraction of cyclical fluctuations in confidence and real macroeconomic variables, especially at medium horizons.
- **Economic takeaway.** Non-fundamental shocks to confidence can be real business-cycle impulses. Sentiment is not only a passive indicator; it can be a causal driver of aggregate fluctuations.

## 2 Incremental contribution of the paper

### 2.1 Contribution relative to the business-cycle shock literature

- The business-cycle literature has extensively identified shocks to monetary policy, fiscal policy, technology, oil prices, and financial conditions.
- Much less direct evidence exists on shocks that are not tied to economic fundamentals, such as animal spirits or autonomous shifts in expectations.
- This paper supplies one of the clearest empirical strategies for isolating a non-fundamental consumer sentiment shock from survey data.

**Incremental contribution:** The paper moves sentiment shocks from a theoretical possibility to a directly measured empirical impulse. Instead of inferring beliefs from macro outcomes, it uses an external source of negative news that plausibly shifts confidence without directly changing fundamentals.

## 2.2 Contribution relative to the consumer confidence literature

- Prior work often documents that consumer confidence forecasts activity or comoves with recessions.
- Forecasting relationships do not identify whether confidence causes activity or merely reflects households' information about fundamentals.
- By instrumenting confidence innovations with mass-shooting fatalities, the paper separates autonomous sentiment from the fundamental information embedded in survey responses.

**Incremental contribution:** The paper turns consumer confidence from a predictive variable into an identified causal object. The key novelty is not using survey data, but extracting the component of survey data unrelated to fundamentals.

## 2.3 Contribution relative to proxy SVAR methodology

- External-instrument SVARs are common in monetary and fiscal policy applications.
- This paper adapts the methodology to a non-policy, non-fundamental shock.
- The authors pay particular attention to weak-instrument inference using Montiel-Olea and Pflueger-style robust statistics and confidence bands.

**Incremental contribution:** The paper demonstrates how proxy SVAR methods can be used to identify belief shocks, not only policy shocks, as long as a credible external source of belief variation is available.

## 2.4 Contribution through the mass-shooting instrument

- The paper builds a harmonized database of mass shootings from 1965 to 2018.
- It focuses on public mass shootings with seven or more fatalities, excluding gang-related and common-place criminal circumstances.
- The instrument is the detrended number of fatalities in such events.
- The authors support the exclusion restriction by showing that local shootings are not predicted by county unemployment and by using individual-level survey responses.

**Incremental contribution:** The paper introduces an unusual but powerful source of variation: tragic public events that are salient, psychologically meaningful, and plausibly orthogonal to macro fundamentals.

## 2.5 Contribution through micro validation

- Macro identification alone could be questioned because aggregate instruments are difficult to validate.
- The paper therefore studies individual survey respondents and county-level exposure to mass shootings.
- Respondents near shootings show a six-month decline in confidence, even after controls for personal finances, county unemployment, time effects, and location clustering.

**Incremental contribution:** The micro evidence closes a key credibility gap. It verifies that the instrument actually moves individual confidence in the expected direction rather than simply appearing relevant in aggregate data.

## 2.6 Contribution through macro findings

- The shock affects real quantities: industrial production, unemployment, labour market tightness, vacancies, and private consumption.
- The shock has weaker effects on prices and nominal variables.
- The pattern suggests a demand-like recessionary disturbance driven by confidence rather than a productivity or inflation shock.

**Incremental contribution:** The paper shows that autonomous confidence shocks have real macroeconomic consequences. It therefore gives empirical support to models in which beliefs, sentiments, or animal spirits generate business-cycle movements.

## 3 Data and measurement

### 3.1 Consumer confidence data

- The main confidence measure is the University of Michigan Index of Consumer Expectations (ICE).
- The ICE aggregates approximately 500 monthly survey responses about expected personal finances and aggregate economic conditions.
- The paper also considers related measures: the Index of Current Economic Conditions (ICC), the overall Index of Consumer Sentiment (ICS), and specific survey questions such as BUS5 and BUS12.
- The ICE is forward looking and is therefore most appropriate for studying expectations and sentiment.
- The index is normalized relative to the 1966 base period and constructed from the difference between positive and negative responses.

**Interpretation:** Consumer confidence is not automatically sentiment. It may combine information about fundamentals with autonomous beliefs. The empirical strategy is designed to isolate the autonomous component.

### 3.2 Survey questions underlying ICE

The ICE is based on questions such as:

1. whether the respondent expects to be better off financially one year from now;
2. whether business conditions will be good or bad over the next twelve months;
3. whether the country will have continuous good times or periods of widespread unemployment or depression over the next five years.

**Interpretation:** These questions are explicitly about expectations. That makes the ICE useful for identifying belief shocks, but it also creates the challenge that beliefs may be informed by fundamentals.

### 3.3 Mass-shooting data

- The paper constructs a new database by harmonizing information from Duwe (2007), Mother Jones (2020), and The Violence Project (2019).
- The benchmark instrument uses public mass shootings with seven or more fatalities.
- The sample excludes shootings in purely private spaces and shootings linked to gang crime or common-place criminal circumstances.
- From January 1965 to November 2018, the database includes 42 mass-shooting events with seven or more fatalities and 581 total fatalities.
- Prominent events include the Columbine High School massacre, the Virginia Tech massacre, the Sandy Hook Elementary massacre, the Orlando nightclub shooting, and the Las Vegas Strip massacre.
- The frequency and severity of mass shootings increase over time, so the paper detrends fatalities using a fourth-order time polynomial.

**Interpretation:** The goal is to capture nationally salient bad news that affects psychological well-being and confidence, while avoiding events whose direct economic consequences might matter.

### 3.4 Instrument construction

The baseline external instrument is:

$$s_t = \text{detrended fatalities in public mass shootings with seven or more fatalities in month } t. \quad (1)$$

Alternative instruments include:

- a dummy for months with a shooting above the cutoff;
- fatalities excluding the Las Vegas shooting;
- instruments based on fatalities or victims;
- longer-sample and shorter-sample variants.

**Interpretation:** Fatalities are used because they summarize the severity and public salience of the event. Dummies are used as robustness checks because extreme fatality counts might otherwise dominate the instrument.

### 3.5 Macroeconomic variables

The benchmark VAR includes:

$$Y_t = [ic_t, u_t, ip_t, cpi_t, ffr_t, unc_t, sp_t], \quad (2)$$

where:

- $ic_t$  is the log of the ICE;
- $u_t$  is the civilian unemployment rate;
- $ip_t$  is log industrial production;
- $cpi_t$  is log CPI;
- $ffr_t$  is the federal funds rate;
- $unc_t$  is log macroeconomic uncertainty from Jurado et al.;
- $sp_t$  is log real stock prices.

**Interpretation:** The benchmark system captures expectations, real activity, labour market conditions, prices, monetary policy, uncertainty, and asset prices.

### 3.6 Additional macro variables

The paper also studies:

- labour market tightness;
- vacancy postings from the Help Wanted index;
- nondurable and durable consumption;
- TFP and utilization-adjusted TFP;
- economic policy uncertainty;
- the VIX.

**Interpretation:** These additional outcomes help determine whether the shock is mainly a labour-market, productivity, uncertainty, or consumption disturbance.

### 3.7 Individual-level survey validation data

- The paper uses individual responses from the Michigan Survey of Consumers.
- It exploits the panel feature that some individuals are interviewed again six months later.
- The main individual outcome is a six-month change in confidence, especially for the BUS5 outlook question.
- Shootings are mapped to counties, and local exposure is measured by county-level fatalities or victims.
- The authors control for changes in personal financial conditions and county unemployment.

**Interpretation:** The micro design tests whether mass shootings directly reduce confidence among individuals geographically exposed to them. This strengthens the interpretation of the aggregate instrument.

## 4 Empirical methodology

### 4.1 Proxy SVAR structure

Let  $Y_t$  be an  $n \times 1$  vector of observables and  $e_t$  be an  $n \times 1$  vector of mutually orthogonal structural shocks. The reduced-form VAR representation is:

$$A(L)Y_t = u_t, \quad (3)$$

where  $A(L) = I - A_1L - A_2L^2 - \dots$ . Reduced-form innovations are linked to structural shocks by:

$$u_t = \Omega_0 e_t. \quad (4)$$

Under stationarity:

$$Y_t = \Psi(L)\Omega_0 e_t. \quad (5)$$

**Interpretation:** The object of interest is the first column of  $\Omega_0$ , which gives the contemporaneous effects of the sentiment shock on all variables.

### 4.2 External-instrument assumptions

Let  $s_t$  be the external instrument and let  $e_{1t}$  denote the sentiment shock. The proxy SVAR assumptions are:

$$E(s_t e_{1t}) = \phi \neq 0, \quad (6)$$

$$E(s_t e_{it}) = 0, \quad i > 1. \quad (7)$$

**Interpretation:** The first condition is relevance: mass-shooting fatalities must move consumer sentiment. The second is exogeneity: the instrument must not be correlated with other macro structural shocks.

### 4.3 Scaling the shock

The impulse responses are scaled so that the sentiment shock produces a one percent decline in the consumer confidence index:

$$\Omega_{0,11} = 1. \quad (8)$$

The remaining coefficients are obtained from:

$$\frac{E(s_t u_{i,t})}{E(s_t u_{1,t})} = \Omega_{0,i1}. \quad (9)$$



**Interpretation:** Scaling makes the magnitude of the shock interpretable: all macro responses are responses to a normalized decline in consumer expectations.

#### 4.4 Implementation

- The VAR includes a constant.
- The benchmark lag length is 18 months, chosen to maximize first-stage relevance.
- Variables other than the federal funds rate are detrended using fourth-order time polynomials.
- The baseline sample is 1965:m1–2007:m8.
- Inference uses weak-instrument robust methods based on Montiel–Olea, Stock, and Watson-style procedures.

**Interpretation:** The lag length and weak-instrument methods are important because the instrument is a rare-event proxy and must be strong enough for reliable inference.

#### 4.5 Local projection IV robustness

The paper also estimates impulse responses using local projections. For horizon  $h$ :

$$y_{i,t+h} - y_{i,t-1} = \alpha_h + \gamma_h ic_t + \varphi_h(L)Y_{t-1} + \varepsilon_{i,t,h}, \quad (10)$$

where  $ic_t$  is instrumented with  $s_t$ .

**Interpretation:** Local projections impose fewer dynamic restrictions than a VAR. Similar results from LP-IV support the robustness of the proxy SVAR findings.

### 5 Identification and credibility checks

#### 5.1 Relevance of the instrument

- Table 1 reports first-stage  $F$ -statistics for the benchmark VAR and alternative confidence measures.
- In the benchmark 1965:m1–2007:m8 sample, the HOM  $F$ -statistic is 11.6 and the robust MOP statistic is 19.3.
- The instrument remains relevant for the longer sample, though weaker after 2007.
- Relevance is also robust to alternative instrument definitions, including shooting dummies and victim counts.

**Interpretation:** The first-stage evidence shows that severe mass-shooting fatalities are predictive of the confidence innovation that the proxy SVAR attributes to sentiment.

#### 5.2 Exogeneity of the instrument

- The exclusion restriction requires mass shootings to be unrelated to macro fundamentals except through sentiment.
- Table 2 shows that lagged county unemployment does not predict the probability of local mass shootings.
- The instrument is based on tragic but rare events whose direct macroeconomic costs are negligible.
- The authors exclude types of shootings more likely to be linked to local economic or criminal conditions.

**Interpretation:** The exogeneity argument is not that mass shootings are random in every respect, but that their timing and severity are not driven by aggregate macro fundamentals in a way that directly affects the VAR outcomes.

### 5.3 Micro evidence for the sentiment channel

- Table 3 regresses six-month changes in individual confidence on local mass-shooting exposure.
- Total fatalities are associated with significant declines in confidence.
- A dummy for large events with at least 10 fatalities also predicts a sharp decline in confidence.
- Results control for changes in personal finances, county unemployment, time fixed effects, and clustered standard errors.

**Interpretation:** This is the key micro validation. It shows that the instrument moves individual sentiment in the same direction as in the aggregate first stage.

### 5.4 Placebo and alternative instrument checks

- Figure 7 uses a shooting dummy instead of fatalities in the benchmark sample.
- Figure 8 repeats the dummy-IV analysis in the longer sample through 2018.
- Figure 9 randomly reshuffles shooting dates to form a placebo instrument.
- Placebo confidence bands become wide and do not produce significant macro effects.

**Interpretation:** These tests show that the estimated effects are not simply generated by the VAR structure or by generic rare-event timing.

## 6 Tables and figures

### 6.1 Figure 1 and Figure 2: confidence comovement and the instrument

**Read:**

- Figure 1 plots ICE against industrial production and unemployment from 1965 to 2018.
- Confidence comoves with the business cycle: it is high in expansions and low in recessions.
- Figure 2 maps public mass shootings with seven or more fatalities and highlights the most lethal events.
- The size of each circle reflects fatalities and the colour scale reflects timing.

**Economic message:** Figure 1 motivates the causal question; Figure 2 shows the external source of variation used to answer it.

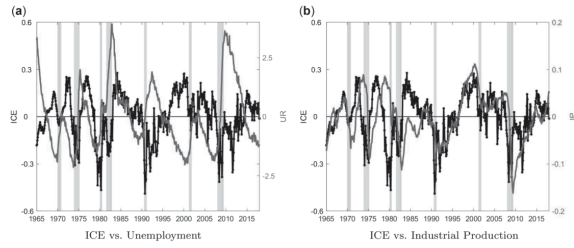


FIGURE 1

Consumer confidence vs. industrial production and unemployment

Notes: The graph presents time series of ICE against industrial production (left panel) and unemployment (right panel) from 1965:1 to 2018:11. All series have been detrended with fourth-order time polynomials. ICE is indicated by lines with circles (black online) using left

Figure 1: ICE vs. unemployment and industrial production

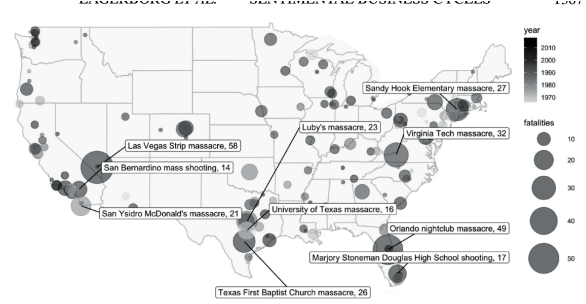


FIGURE 2

Map of mass shootings and fatalities

Figure 2: map of mass shootings and fatalities

## 6.2 Table 1, Table 2, and Table 3: first-stage and micro validation

Read:

- Table 1 shows that mass-shooting fatalities pass relevance tests for the benchmark VAR and alternative confidence specifications.
- Table 2 shows no relationship between lagged county unemployment and the probability of a local shooting.
- Table 3 shows that local shootings reduce individual confidence over six months.

**Economic message:** The three tables support the instrument's credibility: it is relevant for confidence, plausibly exogenous to local economic conditions, and validated at the individual level.

| Part A: Benchmark VAR                                 |                         |                   |                   |
|-------------------------------------------------------|-------------------------|-------------------|-------------------|
| Sample                                                | Proxy                   | F-statistic (HOM) | F-statistic (MOP) |
| 1965:1–2007:8                                         | MassFat                 | 11.6              | 19.3              |
| 1965:1–2018:11                                        | MassFat                 | 10.5              | 5.2               |
| 1965:1–2018:11                                        | MassFat excl. Las Vegas | 14.8              | 8.5               |
| 1965:1–2007:8                                         | MassFat/Dummy           | 12.1              | 17.0              |
| 1965:1–2018:11                                        | MassFat/Dummy           | 8.4               | 5.6               |
| 1965:1–2007:8                                         | MassFat                 | 6.8               | 6.2               |
| 1965:1–2007:8                                         | MassVictims             | 8.9               | 12.9              |
| Part B: Alternative VAR specifications, 1965:1–2007:8 |                         |                   |                   |
| Confidence                                            | Observables             | F-statistic (HOM) | F-statistic (MOP) |
| ICC                                                   | Benchmark               | 5.9               | 8.5               |
| ICS                                                   | Benchmark               | 11.3              | 13.7              |
| BUSS                                                  | Benchmark               | 7.6               | 13.3              |
| BUSS2                                                 | Benchmark               | 7.7               | 11.4              |
| ICE                                                   | CPI inflation           | 10.8              | 16.7              |
| ICE                                                   | no SP500                | 8.8               | 13.1              |
| ICE                                                   | no U12                  | 9.6               | 15.7              |
| ICE                                                   | no SP500, U12           | 7.2               | 10.9              |

Table 1: first-stage relevance

|                             | (1)                      | (2)                      |
|-----------------------------|--------------------------|--------------------------|
| Counties in sample          | ≥ 1 shooting (Fshooting) | All counties (Fshooting) |
| Unemployment rate ( $t-1$ ) | -0.014<br>(0.071)        | -0.001<br>(0.001)        |
| Constant                    | 0.001<br>(0.010)         | 0.000<br>(0.000)         |
| Observations                | 13,020                   | 659,370                  |
| R <sup>2</sup>              | 0.013                    | 0.000                    |
| Number of counties          | 62                       | 3,144                    |
| Time FE                     | ✓                        | ✓                        |
| County FE                   | ✓                        | ✓                        |
| SE cluster                  | State                    | State                    |

Notes: The table reports the estimates of the linear probability model where the probability of mass shootings occurring in a given county is regressed against a constant and the lagged unemployment rate in the county. Standard errors in parentheses \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

Table 2: exogeneity check

|                              | (1)                  | (2)                 | (3)                  | (4)                  | (5)                 |
|------------------------------|----------------------|---------------------|----------------------|----------------------|---------------------|
|                              | Δ Confidence         | Δ Confidence        | Δ Confidence         | Δ Confidence         | Δ Confidence        |
| Total fatalities             | -0.016***<br>(0.007) |                     |                      |                      |                     |
| Total fatalities (min. 7)    |                      | -0.016**<br>(0.007) |                      |                      |                     |
| Dummy for min. 10 fatalities |                      |                     | -0.740***<br>(0.140) |                      |                     |
| Total victims                |                      |                     |                      | -0.008***<br>(0.003) |                     |
| Dummy for min. 10 victims    |                      |                     |                      |                      | -0.430**<br>(0.192) |
| Δ Personal finances          | 0.063***<br>(0.005)  | 0.063***<br>(0.005) | 0.063***<br>(0.005)  | 0.063***<br>(0.005)  | 0.063***<br>(0.005) |
| Δ County unemployment        | -0.010<br>(0.018)    | -0.010<br>(0.018)   | -0.010<br>(0.018)    | -0.010<br>(0.018)    | -0.010<br>(0.018)   |
| Constant                     | 0.153<br>(0.137)     | 0.153<br>(0.137)    | 0.153<br>(0.137)     | 0.153<br>(0.137)     | 0.153<br>(0.137)    |
| Observations                 | 36,276               | 36,276              | 36,276               | 36,276               | 36,276              |
| R <sup>2</sup>               | 0.027                | 0.027               | 0.027                | 0.027                | 0.027               |
| Time FE                      | ✓                    | ✓                   | ✓                    | ✓                    | ✓                   |
| SE cluster                   | MSA                  | MSA                 | MSA                  | MSA                  | MSA                 |
| F-statistic                  | 4.9                  | 5.0                 | 25.0                 | 7.8                  | 5.0                 |

Notes: The table presents estimates on regressing the 6-month change in consumer confidence,  $\Delta_{6C_{i,t}}$ , on county-level mass shootings fatalities accumulated over those 6 months,  $\sum_{j=t-6}^{t-1} N_{j,t}$ . Column (1) presents results for total fatalities;  $F_{shooting}$  (1) presents results for total fatalities;  $F_{shooting}$  (2) presents results for total fatalities;  $F_{shooting}$  (3) presents results for total fatalities;  $F_{shooting}$  (4) presents results for total fatalities;  $F_{shooting}$  (5) presents results for total fatalities.

Table 3: local confidence effect

## 6.3 Figure 3 and Figure 4: first-stage dynamics and identified shocks

Read:

- Figure 3 shows that the ICE declines sharply after a negative instrument shock and recovers gradually over roughly one to two years.
- The response remains robust when individual mass-shooting events are excluded one at a time.
- Figure 4 plots the historical sequence of identified sentiment shocks together with NBER recessions.
- The five-month moving average highlights episodes of sustained negative sentiment.

**Economic message:** The instrument generates a persistent confidence decline and produces historically interpretable sentiment shocks.

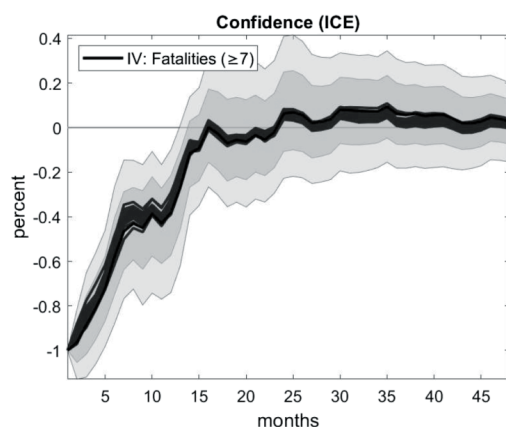


FIGURE 3  
Confidence response to the instrument

*Figure 3: confidence response to instrument*

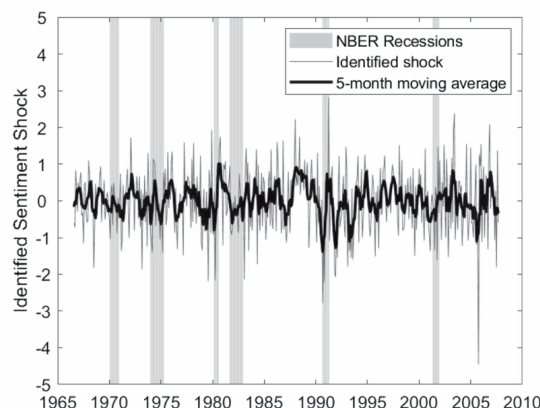


FIGURE 4

Historical realizations of sentiment shocks

ph plots our identified shock series (blue line) and its 5-month moving average (black line) for our benchmark shootings with 7 or more fatalities as an instrument for consumer confidence. Grey shaded areas show NBER

*Figure 4: historical sentiment shocks*

## 6.4 Figure 5 and Figure 6: benchmark macro effects

**Read:**

- Figure 5 shows the benchmark responses of confidence, industrial production, unemployment, CPI, the federal funds rate, uncertainty, and real stock prices.
- A negative sentiment shock lowers confidence for 12–15 months.
- Industrial production falls gradually and persistently.
- Unemployment rises and remains elevated for a long period.
- CPI effects are weak; the federal funds rate declines with a lag; real stock prices fall.
- Figure 6 shows additional real-side effects: labour market tightness and vacancies fall, consumption declines, and TFP is not the main driver.

**Economic message:** Sentiment shocks look like real demand disturbances: they depress real activity and labour markets more clearly than prices.

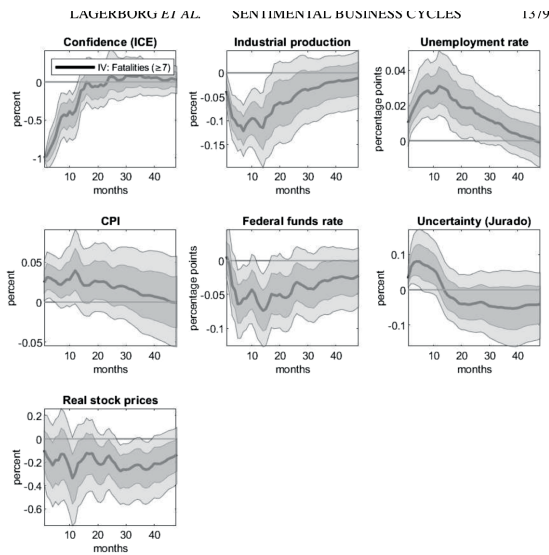


Figure 5: benchmark proxy-SVAR impulse responses

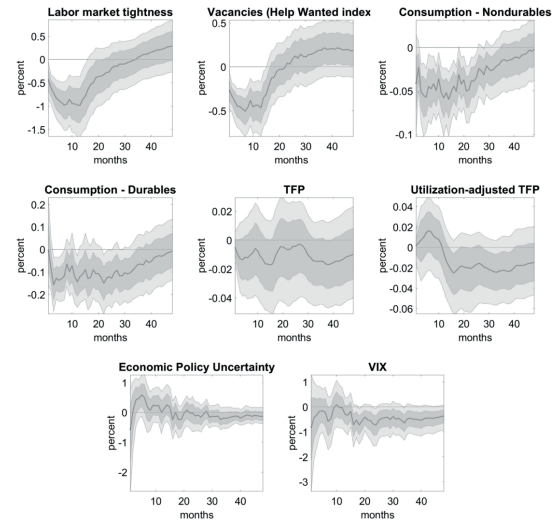


FIGURE 6

Consumer sentiment shock impulse responses—additional variables

Notes: The graph plots impulse response functions to a sentiment shock. The continuous line depicts point estimates of the impact of the

Figure 6: additional variables

## 6.5 Figure 7, Figure 8, and Figure 9: robustness of the instrument

Read:

- Figure 7 uses a shooting dummy instead of fatalities in the benchmark sample and obtains similar dynamics.
- Figure 8 extends the dummy specification to 2018; estimates remain qualitatively similar, but confidence bands widen as the instrument weakens.
- Figure 9 uses a placebo exercise with reshuffled shooting dates. The placebo instrument does not generate significant macro responses.

**Economic message:** The main findings are not driven by one exact fatality measure or by mechanical dynamics in the VAR.

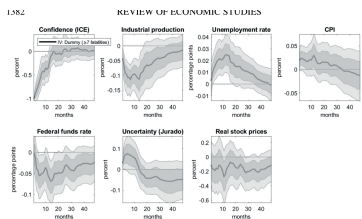


FIGURE 7

Alternative IV: mass shooting dummies, 1965m1–2007m6

Figure 7: dummy IV, benchmark sample

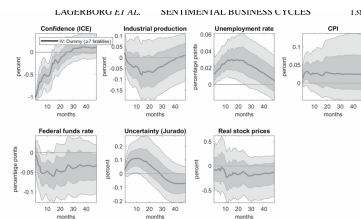


FIGURE 8

Alternative IV: mass shooting dummies, 1965m1–2018m11

Figure 8: dummy IV, long sample

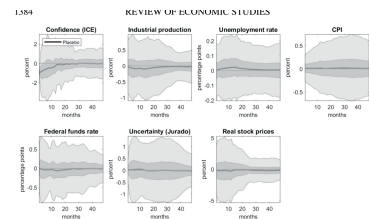


FIGURE 9

Proxy SVAR: placebo with reshuffled shootings

Notes: The graph plots the median with the 90% and 95% estimates at each forecast horizon of the impulse responses over 10,000

Figure 9: placebo IV

## 6.6 Figure 10 and Figure 11: Cholesky comparison

Read:

- Figure 10 identifies a confidence shock using a recursive Cholesky timing assumption instead of the external instrument.
- The Cholesky shock produces broadly similar real effects but also implies an immediate stock-price decline.
- Figure 11 shows that Cholesky identification implies stronger responses of TFP and uncertainty.

**Economic message:** Recursive identification can confound sentiment shocks with productivity news or uncertainty shocks. The proxy SVAR helps isolate the sentiment component.

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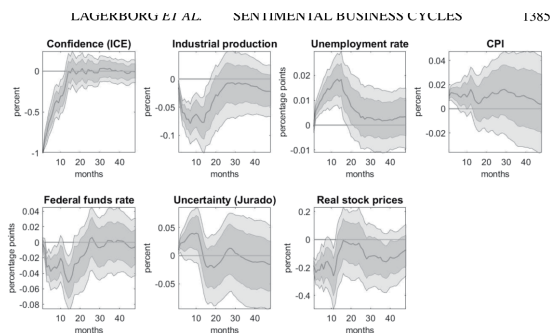


FIGURE 10

Cholesky SVAR—baseline variable responses

Notes: The graph plots the impulse responses to a confidence shock identified by imposing a triangular structure on the covariance matrix.

Figure 10: Cholesky baseline variables

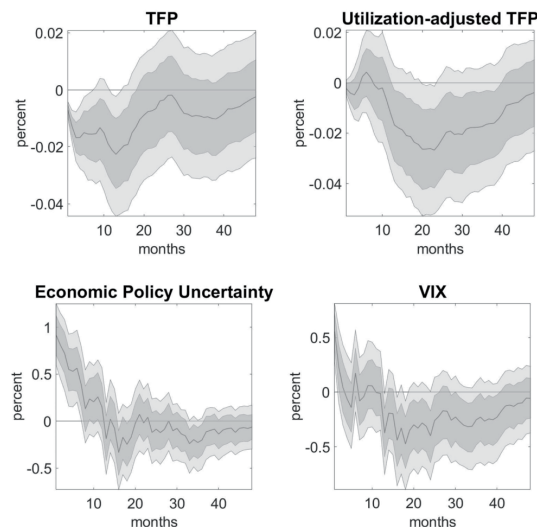


FIGURE 11

Cholesky SVAR—other variables

Figure 11: Cholesky other variables

## 6.7 Figure 12 and Figure 13: local projections and variance contributions

Read:

- Figure 12 shows local-projection IV responses. They are qualitatively similar to the proxy SVAR results, though less precise.
- Confidence declines for about eight months, industrial production falls, and unemployment rises after the shock.
- Figure 13 reports identified sets and confidence bands for forecast variance ratios.
- Sentiment shocks explain a meaningful fraction of variation in ICE and real macro aggregates at medium horizons.

**Economic message:** The LP-IV exercise shows robustness to a different dynamic estimator; the forecast variance results show that sentiment shocks are quantitatively important, not just statistically detectable.

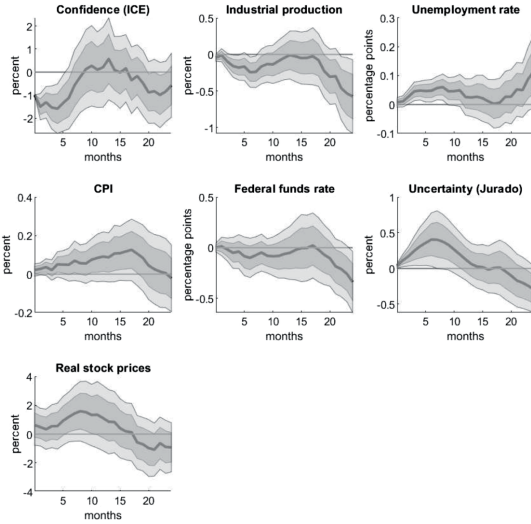


FIGURE 12

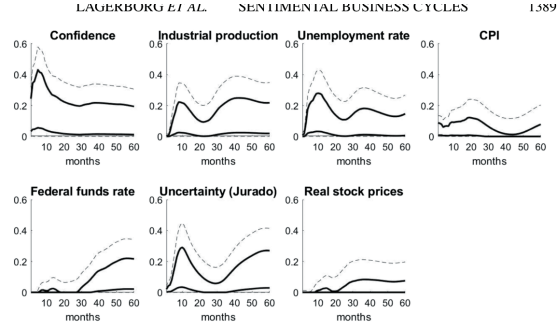
*Figure 12: local projection IV*

FIGURE 13

Identified sets and confidence bands of FVRs

*Figure 13: forecast variance contribution sets*

## 7 Short summary

- The paper estimates the causal macroeconomic effects of autonomous consumer sentiment shocks.
- The main confidence measure is the University of Michigan Index of Consumer Expectations.
- The external instrument is detrended fatalities in public mass shootings with seven or more fatalities.
- The instrument is relevant for confidence and passes weak-instrument tests in the benchmark sample.
- Local micro evidence shows that shootings reduce individual confidence in affected counties.
- A negative sentiment shock lowers confidence persistently.
- Industrial production and private consumption fall; unemployment rises; vacancies and labour-market tightness deteriorate.
- Nominal effects are weaker than real effects.
- The shock is not well explained by TFP, uncertainty, monetary policy, or a Cholesky confidence shock.
- Sentiment shocks explain a non-negligible fraction of cyclical fluctuations in confidence and real activity.

## 8 Conclusion

### 8.1 Core conclusion

Lagerborg, Pappa, and Ravn provide causal evidence that autonomous changes in consumer sentiment can drive business-cycle fluctuations. Using fatalities in mass shootings as an external instrument, they show that negative sentiment shocks trigger persistent declines in confidence and real activity.

### 8.2 Economic intuition

The intuition is that confidence can shift for reasons unrelated to economic fundamentals. When households become pessimistic, they may reduce spending, firms may reduce hiring and production, and labour market conditions deteriorate. These endogenous macro responses validate the idea that beliefs can be independent impulses to the business cycle.

### **8.3 Takeaway**

The paper's main contribution is to identify sentiment shocks directly rather than treating consumer confidence as only a forecasting variable. The findings support macroeconomic models in which animal spirits, expectations, and confidence shocks have real effects.